We are already familiar with the areas of plane figures like square, rectangle, triangle, circle etc. These figures are called 2-dimensional shapes as they can be drawn in a plane. But most of the objects which we come across in our daily life cannot be represented in a plane. For example, tubes, water tanks, bricks, ice-cream cones, football etc. These objects are called solid shapes or 3-dimensional shapes.

We often see solids like cube, cuboid, prism and pyramid. For three dimensional objects measurements like surface area and volume exist.

In this chapter, we shall study about the surface area and volume of some of the standard solid shapes such as cylinder, cone, sphere, hemisphere and frustum.

EZQHPP

7.2 Surface Area

Surface area is the measurement of all exposed area of a solid object.

7.2.1 Right Circular Cylinder

Observe the given figures in Fig.7.2 and identify the shape.

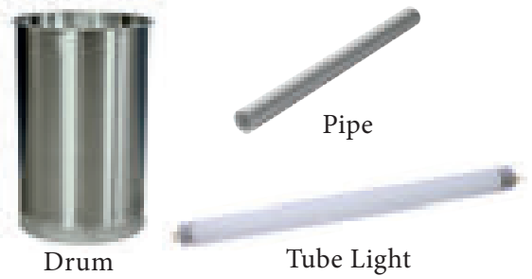


Fig. 7.2

These objects resemble the shape of a cylinder.

Definition : A right circular cylinder is a solid generated by the revolution of a rectangle about one of its sides as axis.

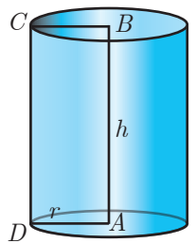


Fig. 7.3

If the axis is perpendicular to the radius then the cylinder is called a right circular cylinder. In the Fig.7.3, AB $= h$ represent the height and $AD = r$ represent the radius of the cylinder.

A solid cylinder is an object bounded by two circular plane surfaces and a curved surface. The area between the two circular bases is called its 'Lateral Surface Area' (L.S.A.) or 'Curved Surface Area' (C.S.A.).

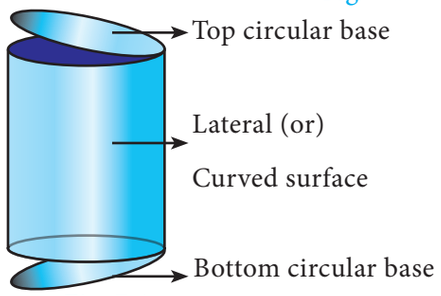


Fig. 7.4

Formation of a Right Circular Cylinder – Demonstration

(i) Take a rectangle sheet of a paper of length l and breadth b.

(ii) Revolve the paper about one of its sides, say b to complete a full rotation (without overlapping).

(iii) The shape thus formed will be a right circular cylinder whose circumference of the base is l and the height is b.

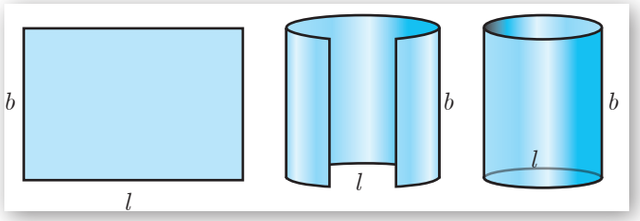


Fig. 7.5

7 Mensuration.indd 270 16-12-2020 03:17:47 PM